Diwali Sales Analysis

Uncovering customer behavior and revenue trends during India's biggest shopping season

PYTHON · PANDAS · NUMPY · MATPLOTLIB

Project Overview

📊 Data Analysis

Analyzed real-world retail sales data captured during the Diwali festival season

🛍️ Customer Behavior

Identified purchasing patterns, preferences, and trends across demographics

📈 Business Impact

Delivered actionable insights to guide marketing and inventory decisions

Dataset at a Glance

Key Features

Demographics

Age, Gender, Marital Status

Geography

State-level transaction data

Transactions

Product category, purchase amount

Real-world transactional records capturing buyer activity across multiple Indian states

Objectives

01

Understand Purchasing Behavior

Map how different customer segments shop during Diwali

02

Identify Top Customers & Products

Pinpoint the highest-value buyers and best-selling categories

03

Analyze Sales Trends

Uncover regional and demographic patterns driving revenue

Data Loading & Exploration

Python Workflow

Import with Pandas

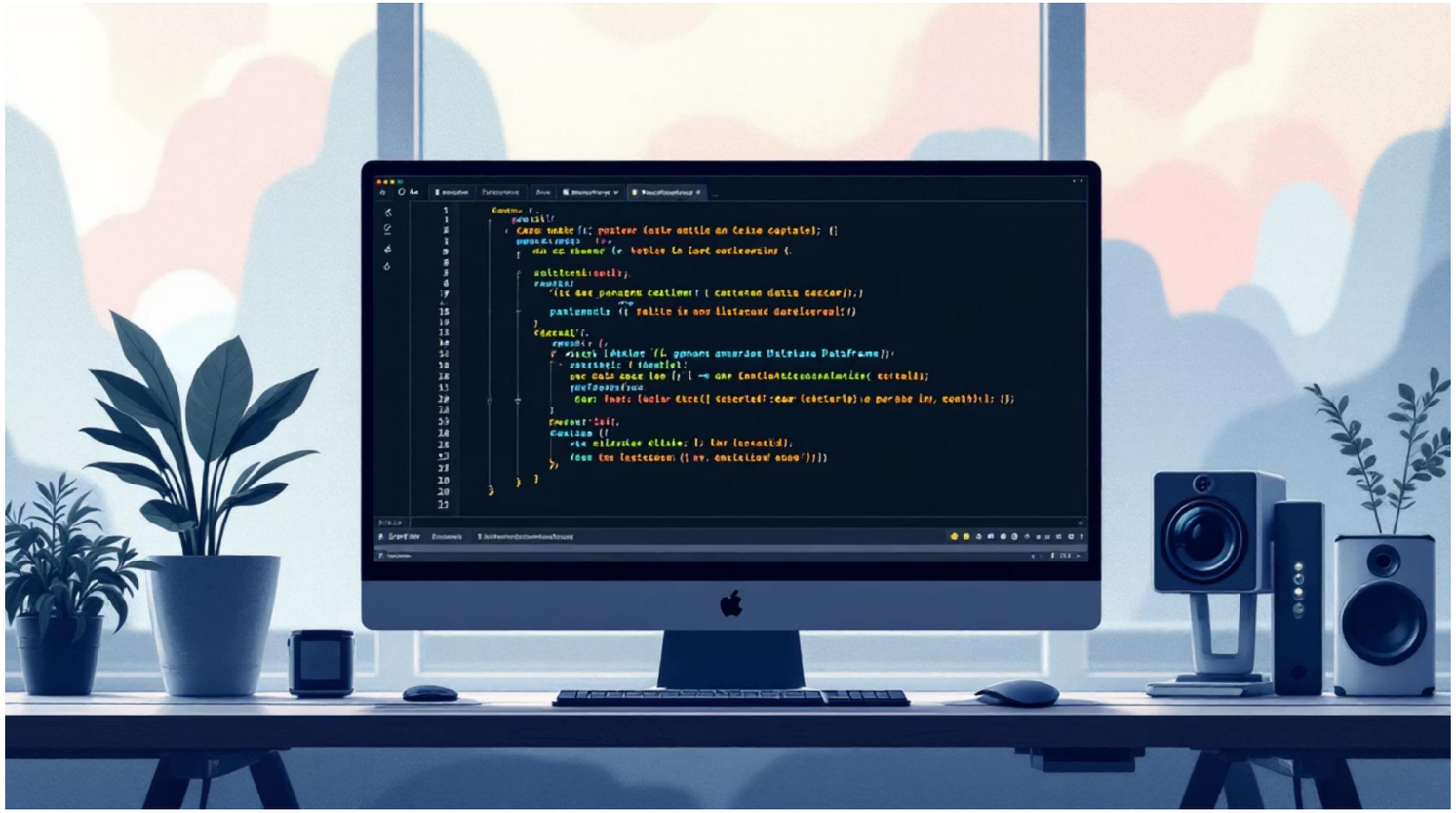
`pd.read_csv()` used to load the dataset into a structured DataFrame

Inspect Structure

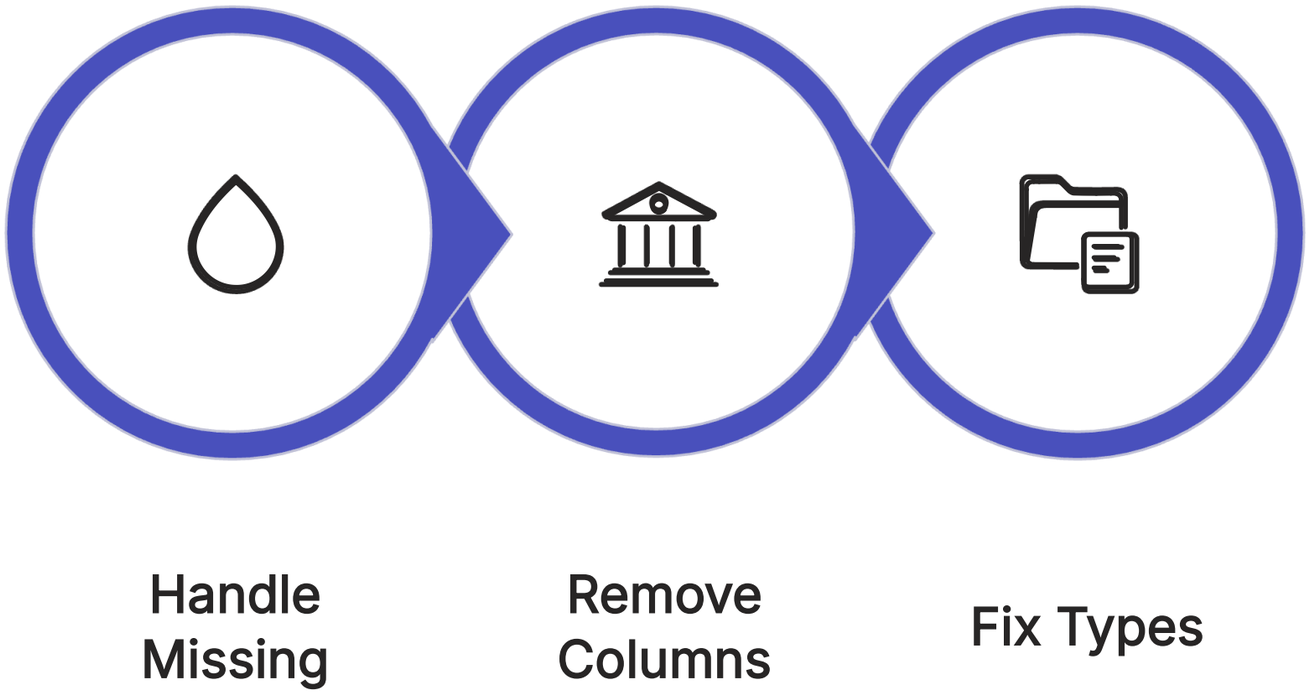
`.info()` revealed column types, nulls, and row count at a glance

Summarize Stats

`.describe()` provided mean, min, max and distribution of numeric fields



Data Cleaning



A clean, well-structured dataset ensures reliable analysis. Each step was performed systematically using Pandas to eliminate noise and prepare data for accurate exploration.

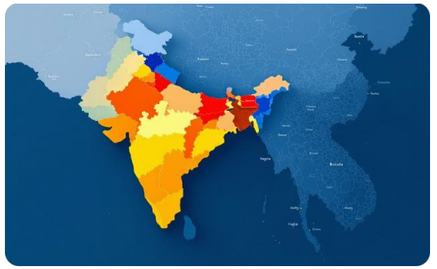
Exploratory Data Analysis

Gender-Wise Sales

Compared total spend and purchase frequency across male and female buyers

Age Group Analysis

Segmented buyers by age to reveal which cohorts drive the most revenue



State-Wise Trends

Mapped regional performance to identify top-contributing states

Key Insights

Women Lead Sales

Female customers contributed the **majority of total revenue** and purchases

Age 26–35 Dominates

This cohort was the **most active buying segment** across all product categories

Top Revenue States

A handful of states generated **disproportionately high** sales volumes

Visualizations

Chart Types Used

Bar Charts

Sales comparisons across gender, age, and state

Count Plots

Frequency distribution of customer segments

Histograms

Purchase amount spread and outlier detection

Built with **Matplotlib** and **Seaborn** for clear, publication-ready visuals

Conclusion & Impact

Target Customer Identified

Female buyers aged 26–35 from top-performing states form the ideal customer profile

Smarter Marketing

Insights enable precise campaign targeting and seasonal budget allocation

Data-Driven Decisions

Replaced guesswork with evidence — a foundation for future festive season strategy

This analysis demonstrates how Python-powered EDA can translate raw transactional data into meaningful business strategy.